

Shravani Mahadeshwar

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Education

09.2025 – 09.2027 (expected)
M.Sc. Applied Data Science — **Frankfurt, Germany**
Frankfurt School of Finance and Management
Grade: 1.9 (German scale) | Coursework: Machine Learning, Deep Learning, AI Ethics, Database and Cloud

09.2021 – 03.2024
B.Sc. Data Science and Business Analytics — **Mumbai, India**
University of Mumbai
Grade: 9.52 / 10.00 | Coursework: NLP, Computer Vision, Pattern Recognition, Time Series Analysis

Skills

Programming	Python, SQL, R, Git
Machine Learning	Explainability (SHAP), Mechanistic Interpretability, RAG Pipelines
Automation & Cloud	n8n, Power Automate, Selenium, REST APIs, AWS (EC2, S3), Terraform, Docker, CI/CD (GitHub Actions)
Data Visualization	Power BI, Tableau, Dashboard Design
Microsoft 365	Excel (Advanced), PowerPoint, SharePoint, Copilot
Languages	English (C1), German (B1)

Professional Experience

06.2024 – 01.2025
Junior Data Science and Business Analyst — **Mumbai, India**
JM Financial Services Limited
- Designed and maintained 4+ enterprise Power BI dashboards by extracting, transforming, and integrating data using SQL from multiple financial databases, enabling executive KPI reporting and client yield analysis.
- Architected a production-ready OCR and signature validation pipeline using OpenCV and EasyOCR with Flask APIs and Dockerized deployment, reducing KYC manual screening by 60%.
- Engineered a Python-Outlook automation framework for recipient-specific report distribution with validation and tracking logic, cutting reporting turnaround time by 70%.
- Queried and analyzed customer data using SQL, applying RFM analysis, feature engineering, and K-Means clustering to deliver data-driven customer segmentation and actionable recommendations.

09.2023 – 02.2024
Data Science Intern — **Mumbai, India**
JM Financial Services Limited
- Built a Selenium-based RPA system for FTP-controlled mutual fund report processing, reducing manual intervention by 90%.
- Developed an NLP pipeline on NIFTY 500 financial news using VADER sentiment analysis to transform unstructured text into structured investment signals.

Projects

03.2026
LLM Audit Bench
- Benchmarked 5 open-source LLMs across trustworthiness dimensions using CrowS-Pairs bias evaluation, perturbation-based robustness testing, SHAP attribution analysis, and privacy-risk assessment.
- Constructed a composite trustworthiness metric, enabling ranked model selection for deployment suitability.

02.2026
Mechanistic Analysis of Transformer Hallucinations
- Investigated factual hallucinations in GPT-Neo-125M through mechanistic interpretability techniques, tracing token propagation across attention heads and residual pathways.
- Developed attribution heatmaps, attention-flow visualisations, and 3D circuit graphs to analyse how erroneous generations emerge within transformer architectures.

10.2025 – 11.2025
Decoding ESG: Measuring Clarity and Greenwashing in Corporate Disclosures
- Conducted a comparative market benchmarking study across 10 years of corporate disclosures from S&P 50 companies, structuring unstructured data into a Python pipeline to support cross-company analysis.
- Designed quantitative benchmarking metrics, including a Greenwashing Risk Index and Concreteness Score, to rank companies on disclosure quality and inform data-driven market intelligence reporting.

Certificates

07.2026 – 08.2026 BlueDot Impact: Technical AI Safety Fellowship

05.2025 – 07.2025 McKinsey Forward Program